WORKED SOLUTION

STEP II 2015, Question 3

Pure • Counting and Induction

> **STEP II 2015, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/15-s2-q3

The key idea

Sort the triangles by their longest rod. The triangles counted by T_n but not by T_{n-1} are exactly those that use the rod of length n, so $T_n - T_{n-1}$ is the number of pairs $a < b < n$ with $a + b \geq n$. For each choice of the middle rod b that is a simple count, and the counts form an arithmetic progression. Adding two consecutive differences gives $T_{2m} - T_{2m-2}$, which is exactly what the inductive step needs.

Solution

Counting the triangles with longest rod c

A triangle counted by T_n has rods $a < b < c \leq n$. Those with $c \leq n-1$ are precisely the triangles counted by T_{n-1}, so

$$T_n - T_{n-1} = \#\{(a,b) : 1 \leq a < b < n,\ a + b \geq n\}.$$

Fix the middle rod b. We need $n - b \leq a \leq b - 1$ (and automatically $a \geq 1$, since $b < n$), which gives $2b - n$ choices when $2b - n > 0$ and none otherwise. So b runs from the least integer above $\frac{n}{2}$ up to $n - 1$.

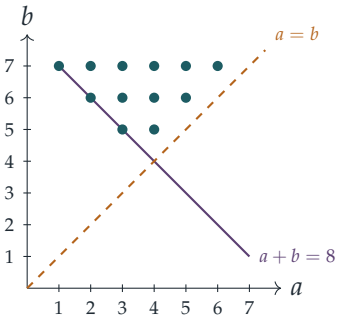
The case $n = 8$

With $c = 8$, the possible middle rods are $b = 5, 6, 7$, giving 2, 4 and 6 choices of a:

$$b = 5 :\ a \in \{3,4\}, \qquad b = 6 :\ a \in \{2,3,4,5\}, \qquad b = 7 :\ a \in \{1,\ldots,6\}.$$

($b = 4$ would need $4 \leq a \leq 3$, which is impossible.) Hence $T_8 - T_7 = 2 + 4 + 6 = 12$, as required.

The picture shows these 12 pairs (a,b): the points on or above the line $a + b = 8$ and strictly above the diagonal $a = b$.



For $T_7 - T_6$, the longest rod is 7 and the middle rod $b = 4,5,6$ gives $2b - 7 = 1,3,5$ choices, so $T_7 - T_6 = 1 + 3 + 5 = 9$. Therefore

$$T_8 - T_6 = (T_8 - T_7) + (T_7 - T_6) = 12 + 9 = 21.$$

The general differences

For $n = 2m$, the middle rod runs over $b = m+1, \ldots, 2m-1$, with $2b - 2m = 2, 4, \ldots, 2(m-1)$ choices:

$$T_{2m} - T_{2m-1} = 2 + 4 + \cdots + 2(m-1) = m(m-1).$$

For $n = 2m - 1$, the middle rod runs over $b = m, \ldots, 2m-2$, with $2b - (2m-1) = 1, 3, \ldots, 2m-3$ choices, and the sum of the first $m-1$ odd numbers is $(m-1)^2$. Adding,

$$T_{2m} - T_{2m-2} = m(m-1) + (m-1)^2 = (m-1)(2m-1).$$

(With $m = 4$ these give 12 and 21, matching the values above.) These hold whenever the smaller index is at least 3, i.e. for $m \geq 3$ in the second formula.

The induction

Let $f(m) = \frac{1}{6}m(m-1)(4m+1)$. We prove $T_{2m} = f(m)$ for all $m \geq 2$.

Base case, $m = 2$. From the rods $1,2,3,4$ the four possible choices are $(1,2,3)$, $(1,2,4)$, $(1,3,4)$, $(2,3,4)$, and all except $(1,2,4)$ satisfy $a + b \geq c$. So $T_4 = 3$, and $f(2) = \frac{1}{6} \cdot 2 \cdot 1 \cdot 9 = 3$.

Inductive step. Suppose $T_{2k-2} = f(k-1)$ for some $k \geq 3$. By the difference formula,

$$T_{2k} = f(k-1) + (k-1)(2k-1),$$

so it is enough to show $f(k) - f(k-1) = (k-1)(2k-1)$.

Ask yourself How do we tidy up $f(k) - f(k-1)$ without expanding two cubics in full?

Both terms contain the factor $k-1$, so take it out first:

$$\begin{aligned} f(k)-f(k-1) &= \tfrac{1}{6}(k-1)\big[k(4k+1)-(k-2)(4k-3)\big] \\ &= \tfrac{1}{6}(k-1)\big[(4k^2+k)-(4k^2-11k+6)\big] \\ &= \tfrac{1}{6}(k-1)(12k-6) = (k-1)(2k-1). \end{aligned}$$

Hence $T_{2k} = f(k)$. By induction, $T_{2m} = \tfrac{1}{6}m(m-1)(4m+1)$ for every $m \geq 2$.

An odd number of rods

Using $T_{2m-1} = T_{2m} - m(m-1)$,

$$T_{2m-1} = \tfrac{1}{6}m(m-1)(4m+1) - m(m-1) = \tfrac{1}{6}m(m-1)(4m-5) \qquad (m \geq 2).$$

Writing the number of rods as $2m+1$ instead (replace m by $m+1$), this is

$$T_{2m+1} = \tfrac{1}{6}m(m+1)(4m-1) \qquad (m \geq 1).$$

For example $T_3 = \tfrac{1}{6}\cdot 1\cdot 2\cdot 3 = 1$, the single triangle $(1,2,3)$.

Checks

- The formulas give $T_6 = \tfrac{1}{6}\cdot 3\cdot 2\cdot 13 = 13$, $T_7 = \tfrac{1}{6}\cdot 3\cdot 4\cdot 11 = 22$ and $T_8 = \tfrac{1}{6}\cdot 4\cdot 3\cdot 17 = 34$, so $T_8 - T_7 = 12$ and $T_8 - T_6 = 21$, as found directly.
- A computer count of all triples from $\{1,\ldots,n\}$ agrees with both closed forms for every n from 3 to 79.
- With $n = 2m$, the even formula is about $\tfrac{2}{3}m^3 = \tfrac{n^3}{12}$, and the odd one behaves the same way. That is half of the $\binom{n}{3} \approx \tfrac{n^3}{6}$ possible triples, so for large n about half of all choices of three rods make a triangle.

Common slips

- Forgetting that the rods are distinct, so $a < b$ strictly: allowing $a = b$ overcounts.
- Losing the zero-area (degenerate) triangles: the condition is $a + b \geq c$, so $(1,2,3)$ counts.
- Starting the induction at $m = 1$: T_2 is not defined, since we need $n \geq 3$.

Beyond STEP

Counting lattice points in a region, as in the picture, is the start of the theory of partitions and generating functions: T_n has a rational generating function, and the split into even and odd n reflects a factor $1 - x^2$ in its denominator. Results like this, where

a count is a polynomial in n that depends on n's residue, are called quasi-polynomials. For more on recurrences and counting arguments, see Siklos, *Advanced Problems in Mathematics*.

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